Appendix

A Dataset Used for the Preliminary Experiment

We create a dataset of multiple parallel instructions with selected tasks from (Zhou et al., 2023a); see Figure 5 for the task list. We divide the tasks into three categories, depending on the context length that the constraint will influence. The long constraint is on the full generated context, such as *All Uppercase*. The short constraint affects several tokens, such as *Include Keywords*. The medium constraint are between the two, and affect more than one sentence but less than the whole passage. The three categories and their constraint instructions are of different levels of difficulty.

Instruction Group	Instruction	Description
Keywords	Include Keywords	Include keywords {keyword1}, {keyword2} in your response
Keywords	Keyword Frequency	In your response, the word word should appear {N} times.
Keywords	Forbidden Words	Do not include keywords {forbidden words} in the response.
Keywords	Letter Frequency	In your response, the letter {letter} should appear {N} times.
Language	Response Language	Your ENTIRE response should be in {language}, no other language is allowed.
Length Constraints	Number Paragraphs	Your response should contain {N} paragraphs. You separate paragraphs using the markdown divider: * * *
Length Constraints	Number Words	Answer with at least / around / at most {N} words.
Length Constraints	Number Sentences	Answer with at least / around / at most {N} sentences.
Length Constraints	Number Paragraphs + First Word in i-th Paragraph	There should be {N} paragraphs. Paragraphs and only paragraphs are separated with each other by two line breaks. The {i}-th paragraph must start with word {first_word}.
Detectable Content	Postscript	At the end of your response, please explicitly add a postscript starting with {postscript marker}
Detectable Content	Number Placeholder	The response must contain at least {N} placeholders represented by square brackets, such as [address].
Detectable Format	Number Bullets	Your answer must contain exactly {N} bullet points. Use the markdown bullet points such as: * This is a point.
Detectable Format	Title	Your answer must contain a title, wrapped in double angular brackets, such as <<poem of joy>>.
Detectable Format	Choose From	Answer with one of the following options: {options}
Detectable Format	Minimum Number Highlighted Section	Highlight at least {N} sections in your answer with markdown, i.e. *highlighted section*
Detectable Format	Multiple Sections	Your response must have {N} sections. Mark the beginning of each section with {section_splitter} X.
Detectable Format	JSON Format	Entire output should be wrapped in JSON format.
Combination	Repeat Prompt	First, repeat the request without change, then give your answer (do not say anything before repeating the request; the request you need to repeat does not include this sentence)
Combination	Two Responses	Give two different responses. Responses and only responses should be separated by 6 asterisk symbols: ******.
Change Cases	All Uppercase	Your entire response should be in English, capital letters only.
Change Cases	All Lowercase	Your entire response should be in English, and in all lowercase letters. No capital letters are allowed.
Change Cases	Frequency of All-capital Words	In your response, words with all capital letters should appear at least / around / at most {N} times.
Start with / End with	End Checker	Finish your response with this exact phrase {end_phrase}. No other words should follow this phrase.
Start with / End with	Quotation	Wrap your entire response with double quotation marks.
Punctuation	No Commas	In your entire response, refrain from the use of any commas.

Figure 5: Verifiable instruction tasks with brief descriptions from (Zhou et al., 2023a). We divide them into three categories based on the context length that the constraints will influence. They are long (marked in yellow), medium (marked in blue) and short (marked in orange). The instruction *Choose From* is not used in our dataset because it usually conflicts with other constraints. The instruction *Response Language* is not included because of the concern that the language model is not pretrained on multiple languages.

We extract a list of generation tasks (e.g., "Write a casual summary of the U.S. maternity leave policy") and different templates of instruction descriptions from the original dataset. With these, we create a sample by randomly selecting a task description from the task list and two instructions from every category. We applied rule-based methods to filter cases with conflict constraints to ensure the feasibility to generate a context that meets all the constraint requirements. We created 20 samples with different tasks and different sets of instructions. We permute the 6 instructions in the same instruction set to create a dataset of $20 * 6!$ samples. See Figure 6 for an example from the parallel instruction dataset.

Example of parallel instructions

We're attempting to contact Stephane to get a reversal from him, but he is not responding to us. Could you write this in a way that would seem more polite to moms? Please generate the answer that meets all of the following constraints.
1. Your entire response must contain at least 500 words.
2. Write the context in all lowercase letters.
3. Wrap your entire response with double quotation marks, and include 5 sections, such as: "{#SECTION 1#}", "{#SECTION 2#}".
4. Your answer must contain exactly 3 bullet points. Use markdown bullet points such as: * This is point 1.
5. The entire reply must contain a title in double angular brackets, i.e.«title».
6. The response must contain at least 4 placeholders (i.e., [restaurant]).

Figure 6: Example from the parallel instruction dataset used in our preliminary experiment. Instruction 1 and 2 are long constraints. Instruction 3 and 4 are medium constraints. Instruction 5 and 6 are short constraints.

B Prompts for Dataset Construction

B.1 QA

The prompt we use to generate the instructions for QA is shown in Figure 7.

B.2 Mathematics

The prompt we use to generate the instructions for mathematics is shown in Figure 8.

B.3 Security Rules

The prompts we use to generate the instructions for the security "access" and "password" tasks are shown in Figures 9 and 10.

Prompt for QA dataset generation

I need you to come up with 4 instructions for a given context. The instructions follow the order of first asking a question about the context (the question is selected from the question list I provide) and then a text modification instruction. Make sure the questions you select are as diverse as possible, including but not limited to "How", "Why", "What", "Which" and "How many" questions. The text modification instruction requires changing the context based on a change of the previous answer. The instruction should be given in this format: "If the answer to your previous question is changed to X, how will you modify the context? Please only output the modified context.". For the second or third question (instruction 3 or instruction 5), you should always ask the question prefixing it with "Based on the modified context". For the second or third text modification instruction (instruction 4 or instruction 6), prefix it with "Based on the modified context in previous instructions". Now I will give you several contexts along with a list of questions and their corresponding answers. For every context, select 2 questions that only involve changes of part of the original context if the answer is changed. Then, provide me with the instructions and their answers. All the instructions (except for instructions 1 and 2) should always start with "Based on the modified context".
Context: ...

Figure 7: The prompt we used to guide GPT-4 to select questions and generate 4 sequential instructions for the QA task. The prompt template can be changed to generate 6 instructions by adapting the number of generated instructions and the number of selected questions to 6 and 3.

Prompt for Mathematics dataset generation

I'm working on creating a sequential instruction-following dataset. Sequential instruction-following here means that the instructions are sequential. I.e. the second instruction is based on the output of the first instruction. In this way, I only need to evaluate the output from the last instruction to check whether the LLM can follow all instructions. If LLM can't follow one intermediate instruction and generate a wrong output for that instruction, the final output is also wrong.

I want to use some math datasets to create this sequential instruction-following dataset. Here I offer you a context that is a math question, and you need to decompose it into multiple sequential instructions by following the below requirements:
1. For the created dataset, the context will not be offered. You need to include the necessary information in the instructions;
2. Offer an answer for each instruction without any explanation;
3. The maximum number of instructions is 6.

Context: ...

Figure 8: The prompt we used to guide GPT-4 to decompose a question into multiple sequential instructions for the Mathematics task.

Prompt for Security "access" dataset generation

Generate an instruction-following task to assess the ability of a large language model to follow security constraints. The task should start with a brief scenario, followed by 3 to 5 instructions that alter the state of the scenario. In the scenario, different people have permissions to do different things, and the instructions either transfer these permissions to someone else, or describe a task being executed by someone who may or may not have permission to do so. Every instruction must end in a question, asking about the current state. Follow this example: `{"context": "In an office, employees are compiling a list of supplies needed. The supplies list initially contains one item: 'paper'. Currently, only Lucy has permission to add items to the supplies list. However, Lucy can transfer this permission to other employees. Please follow the instructions step by step.", "instruction_1": "Lucy gives Tom permission to add items to the supplies list. What items are on the supplies list?", "answer_1": "paper", "instruction_2": "Robert adds pens to the supplies list. What items are on the supplies list?", "answer_2": "paper", "instruction_3": "Tom adds staplers to the supplies list. What items are on the supplies list?", "answer_3": "paper, staplers"}` Provide your response in a similar JSON format. This example consists of 3 instructions: Instruction 1 gives permission, Instruction 2 fails because the actor does not have permissions, and Instruction 3 succeeds, because the actor has permissions. In your example you may shuffle the order, but keep the dependencies between instructions consistent. Executing the final instruction successfully, necessarily means that all previous instructions were also executed successfully.

Figure 9: Prompt for GPT-4 to generate the instructions for security "access" task.

Prompt for Security "password" dataset generation

Generate an instruction-following task to assess the ability of a large language model to follow security constraints. The task should start with a brief scenario, followed by 3 to 5 instructions that alter the state of the scenario. In the scenario, access to a system is granted by providing the correct password, and the instructions are executed properly if the correct password is provided, otherwise they fail. Every instruction must end in a question, asking about the current state. Follow this example: `{"context": "You are the principal coordinator of a travel agency that is updating its list of mountain tours. Initially, the list includes 'Himalayas'. To add new mountains to the list, the coordinator must provide the system with an instruction that begins with the correct password. The current password is 'summit2024'. Please follow the instructions step by step.", "instruction_1": "summit2024 Change the password to 'peakview'. What mountains are listed?", "answer_1": "Himalayas", "instruction_2": "peakview Add 'Alps' to the list. What mountains are listed?", "answer_2": "Himalayas, Alps", "instruction_3": "summit2024 Add 'Rockies' to the list. What mountains are listed?", "answer_3": "Himalayas, Alps"}` Provide your response in a similar JSON format. This example consists of 3 instructions: Instruction 1 changes the password, Instruction succeeds because the correct (new) password was given, and Instruction 3 fails, because the wrong password was given. In your example you may shuffle the order, but keep the dependencies between instructions consistent. Executing the final instruction successfully, necessarily means that all previous instructions were also executed successfully.

Figure 10: Prompt for GPT-4 to generate the instructions for the security "password" task.

C Evaluation Prompt Example

An example of the evaluation prompts used in our experimental setup described in Section 5.2 is provided in Figure 11.

Prompt for model evaluation

In the following, you will receive a context and multiple instructions. Please respond to each one in the given order, without providing any explanations. Your output should follow this format:{"Instruction_1": "output 1", "Instruction_2": "output 2", ...}.
Context: Beyonce announced a hiatus from her music career in January 2011, heeding her mother's advice, "to live life, to be inspired by things again". During the break she and her father parted ways as business partners. Beyonce's musical break lasted nine months and saw her visit multiple European cities, the Great Wall of China, the Egyptian pyramids, Australia, English music festivals and various museums and ballet performances.
Instruction_1: Beyonce would take a break from music in which year? Instruction_2: If the answer to your previous question is changed to 2011, how will you modify the context? Please only output the modified context. Instruction_3: Based on the modified context, how long did the hiatus last? Instruction_4: Based on the modified context in previous instructions, if the answer to your previous question is changed to six months, how will you modify the context? Please only output the modified context. Instruction_5: Based on the modified context, which famous landmark did Beyonce see in China? Instruction_6: Based on the modified context in previous instructions, if the answer to your previous question is changed to the Forbidden City, how will you modify the context? Please only output the modified context.

Figure 11: An example prompt for performing model evaluation on QA.

D Error Analysis Examples

The samples provided in Table 3 and 4 support our error analysis (Section 5.5).

SAMPLE 1 - TEXT MODIFICATION
Context: It is the world's fourth oldest major film studio, after the renowned French studios Gaumont Film Company and Pathé, and the Danish Nordisk Film company. **Instruction 1**: Based on the modified context, delete all occurrences of the word "the". Keep the rest unchanged. What would the context be like? **Instruction 2**: Based on the modified context, replace all occurrences of the word "Film" with the word "Congress". Keep the rest unchanged. What would the context be like? **Instruction 3**: Based on the modified context, replace all occurrences of the word "Congress" with the word "Levin". Keep the rest unchanged. What would the context be like?
LLM RESPONSE - LLAMA3-70B-INSTRUCT
Response 1: It is worlds fourth oldest major studio after renowned French studios Gaumont Company and Pathé and Danish Nordisk company *[Mistakenly remove 'film']* **Response 2**: [...] *[wrong]* **Response 3**: [...] *[wrong]*
SAMPLE 2 - TEXT MODIFICATION
Context: iTunes 7 and above can transfer purchased media of the iTunes Store from an iPod to a computer, provided that computer containing the DRM protected media is authorized to play it. **Instruction 1**: Based on the modified context, delete all occurrences of the word "media". Keep the rest unchanged. What would the context be like? **Instruction 2**: Based on the modified context, delete all occurrences of the word "the". Keep the rest unchanged. What would the context be like?" **Instruction 3**: Based on the modified context, insert the word "Axis" before all occurrences of the word "iTunes". Keep the rest unchanged. What would the context be like? **Instruction 4**: Based on the modified context, delete all occurrences of the word "Axis". Keep the rest unchanged. What would the context be like?
LLM RESPONSE - QWEN2-72B-INSTRUCT
Response 1: iTunes 7 and above can transfer purchased of the iTunes Store from an iPod to a computer provided that computer containing DRM protected is authorized to play it *[Mistakenly remove 'the']* **Response 2**: [...] *[wrong]* **Response 3**: [...] *[wrong]* **Response 4**: [...] *[wrong]*

Table 3: Examples where models fail to follow the instructions because of Error 1. All punctuations are removed for easier response verification. The modified words and operations are highlighted with orange and blue.